

4 Safety Engineering Quiz ANS

1. In the two-factor risk equation, risk is commonly decomposed into which factors?

- A) Probability and consequence
- B) Speed and memory
- C) Precision and recall
- D) Cost and revenue only

ANS: A) Probability and consequence

2. What are 'nines of reliability' used to express?

- A) The number of system components
- B) The proportion of time or trials in which a system performs successfully
- C) The severity of one failure
- D) The length of a test plan

ANS: B) The proportion of time or trials in which a system performs successfully

3. What is redundancy?

- A) Giving one component unlimited authority
- B) Removing backups
- C) Using multiple components that can perform the same crucial function
- D) Combining every duty into one agent

ANS: C) Using multiple components that can perform the same crucial function

4. What is the principle of least privilege?

- A) Every agent should have administrator access
- B) All agents should share every credential
- C) Privileges should grow without limit
- D) Each agent should receive only the minimum power needed for its task

ANS: D) Each agent should receive only the minimum power needed for its task

5. What does separation of duties seek to prevent?

- A) A single agent controlling an entire critical process
- B) Specialization among agents
- C) Independent checks
- D) Division of responsibilities

ANS: A) A single agent controlling an entire critical process

6. What is a fail-safe?

- A) A component that always increases output
- B) A feature designed to leave the system safe when failure occurs
- C) A method for hiding errors
- D) A system with no shutdown state

ANS: B) A feature designed to leave the system safe when failure occurs

7. What is antifragility?

- A) Resistance to all change
- B) The absence of stress
- C) The capacity to improve or become stronger through adversity
- D) Reliance on one fragile component

ANS: C) The capacity to improve or become stronger through adversity

8. What is defense in depth?

- A) One perfect security barrier
- B) No separation between layers
- C) Trusting every input
- D) Multiple protective layers so one failure does not defeat the entire defense

ANS: D) Multiple protective layers so one failure does not defeat the entire defense

9. What is the Swiss Cheese Model intended to illustrate?

- A) Accidents occur when weaknesses in multiple defensive layers align
- B) All accidents have one simple cause
- C) Human factors never matter
- D) Every layer is flawless

ANS: A) Accidents occur when weaknesses in multiple defensive layers align

10. What does Fault Tree Analysis begin with?

- A) A random success
- B) An undesirable top event and the combinations of failures that can cause it
- C) Only organizational culture
- D) A list of employee names

ANS: B) An undesirable top event and the combinations of failures that can cause it

11. Why are component-failure models limited for complex systems?

- A) Components never fail
- B) They contain too much sociology
- C) Accidents may emerge from interactions even when individual components work as designed
- D) They prohibit logical analysis

ANS: C) Accidents may emerge from interactions even when individual components work as designed

12. What does 'drift into failure' emphasize?

- A) Catastrophe is always intentional
- B) Only hardware matters
- C) Systems remain static
- D) Normal adaptations and pressures can gradually move a system toward unsafe operation

ANS: D) Normal adaptations and pressures can gradually move a system toward unsafe operation

13. What does Normal Accident Theory associate with unavoidable accidents?

- A) Interactive complexity and tight coupling
- B) Complete component independence
- C) Perfect transparency
- D) Only low-risk environments

ANS: A) Interactive complexity and tight coupling

14. How do High Reliability Organizations respond to weak warning signals?

- A) They ignore them
- B) They remain preoccupied with failure and study anomalies and near misses
- C) They punish all reporting
- D) They assume past success guarantees future safety

ANS: B) They remain preoccupied with failure and study anomalies and near misses

15. In STAMP, safety is treated primarily as what?

- A) A property of one component
- B) A purely statistical average
- C) An emergent system property maintained through control constraints
- D) A synonym for reliability only

ANS: C) An emergent system property maintained through control constraints

16. What is a tail event?

- A) A routine median outcome
- B) An event that changes no risk estimate
- C) A guaranteed event
- D) A rare event located in the extreme part of a probability distribution

ANS: D) A rare event located in the extreme part of a probability distribution

17. What characterizes Extremistan?

- A)** Rare extreme events can dominate totals and averages
- B)** Outcomes remain tightly bounded
- C)** The mean is unaffected by outliers
- D)** Only thin-tailed distributions occur

ANS: A) Rare extreme events can dominate totals and averages

18. What is a Black Swan?

- A)** A frequent, fully predictable event
- B)** A rare, high-impact event that is difficult to anticipate from past observations
- C)** A harmless known failure
- D)** A median observation

ANS: B) A rare, high-impact event that is difficult to anticipate from past observations

19. What lesson does the turkey problem illustrate?

- A)** Past survival proves permanent safety
- B)** More data always eliminates uncertainty
- C)** A long record without disaster may not reveal an abrupt catastrophic risk
- D)** Tail risks can be ignored

ANS: C) A long record without disaster may not reveal an abrupt catastrophic risk

20. Which statement is important when evaluating unobserved AI hazards?

- A)** No incidents proves no risk
- B)** Successful safety measures are unnecessary
- C)** Only average outcomes matter
- D)** Absence of evidence is not strong evidence of absence

ANS: D) Absence of evidence is not strong evidence of absence